Math Refresher for Masters

Homework 1.

1 $x = (x_1, x_2) \in \mathbb{R}^2$

$$\|x\|_1 = |x_1| + |x_2|$$
$$\|x\|_2 = \sqrt{x_1^2 + x_2^2}$$
$$\|x\|_\infty = \max\{|x_1|, |x_2|\}$$

A) 1) $|x_1| \geq 0$ and $|x_2| \geq 0 \Rightarrow \boxed{\|x\|_1 \geq 0}$

if $\|x\|_1 = 0$, then $|x_1| + |x_2| = 0 \Rightarrow x_1 = x_2 = 0 \Rightarrow \boxed{x = (0,0) \Leftrightarrow \|x\|_1 = 0}$

2) $\|\alpha x\|_1 = |\alpha x_1| + |\alpha x_2|$ $\qquad |\alpha x_i| = |\alpha| \cdot |x_i|$

$$|\alpha| \cdot |x_1| + |\alpha| \cdot |x_2| = |\alpha| \cdot (|x_1| + |x_2|) \Rightarrow \boxed{\|\alpha x\|_1 = |\alpha| \cdot \|x\|_1}$$

3) Let's $x = (x_1, x_2)$, $y = (y_1, y_2)$, then

$\|x+y\|_1 = |x_1 + y_1| + |x_2 + y_2|$ $\qquad |a+b| \leq |a| + |b|$

$|x_1 + y_1| \leq |x_1| + |y_1|$ $\qquad |x_2 + y_2| \leq |x_2| + |y_2|$

$\boxed{\|x+y\|_1 \leq} |x_1| + |y_1| + |x_2| + |y_2| = \boxed{\|x\|_1 + \|y\|_1}$ $\Rightarrow$ all three conditions are correct.

B) 1) $x_1^2 \geq 0$, $x_2^2 \geq 0 \Rightarrow \boxed{\|x\|_2 \geq 0}$

if $\|x\|_2 = 0$, then $\sqrt{x_1^2 + x_2^2} = 0 \Rightarrow x_1^2 + x_2^2 = 0 \Rightarrow \boxed{x_1 = x_2 = 0}$

2) $\boxed{\|\alpha x\|_2} = \sqrt{(\alpha x_1)^2 + (\alpha x_2)^2} = \sqrt{\alpha^2 (x_1^2 + x_2^2)} = |\alpha| \sqrt{x_1^2 + x_2^2} = \boxed{|\alpha| \cdot \|x\|_2}$

3) $\boxed{\|x+y\|_2^2} = (x_1 + y_1)^2 + (x_2 + y_2)^2 = x_1^2 + 2x_1y_1 + y_1^2 + x_2^2 + 2x_2y_2 + y_2^2 =$

$= x_1^2 + x_2^2 + y_1^2 + y_2^2 + 2(x_1y_1 + x_2y_2) \ominus$

using th. Cauchy-Schwarz: $x_1y_1 + x_2y_2 \leq \sqrt{x_1^2 + x_2^2}\sqrt{y_1^2 + y_2^2} \Rightarrow$

$\ominus (x_1^2 + x_2^2) + 2\sqrt{x_1^2 + x^2}\sqrt{y_1^2 + y_2^2} + (y_1^2 + y_2^2) = \left(\sqrt{x_1^2 + x_2^2} + \sqrt{y_1^2 + y_2^2}\right)^2 \boxed{= \left(\|x\|_2 + \|y\|_2\right)^2} \Rightarrow$ it's good!

3 $P_n = \{\text{real polynomials of degree} \leq n\}$ on $[0;1]$

(a) $\langle p, q \rangle = \int_0^1 p(x)q(x)\,dx$

1) $p(x)q(x) = q(x)p(x) \Rightarrow \boxed{\langle q, p \rangle = \int_0^1 q(x)p(x)\,dx = \int_0^1 p(x)q(x)\,dx = \langle p, q \rangle}$

2) $\boxed{\langle \alpha p + \beta r, q \rangle} = \int_0^1 (\alpha p(x) + \beta r(x))\,q(x)\,dx = \int_0^1 \alpha p(x)q(x)\,dx + \int_0^1 \beta r(x)q(x)\,dx = \boxed{\alpha\langle p, q \rangle + \beta\langle r, q \rangle}$

3) $\langle p, p \rangle = \int_0^1 p(x)p(x)\,dx = \int_0^1 p(x)^2\,dx$

if $\int_0^1 p(x)^2\,dx = 0 \Rightarrow \boxed{p(x) = 0}$ using continuity and $p(x)^2 \geq 0$.

(b) $\|p\|_2 = \sqrt{\int_0^1 p(x)^2\,dx}$

1) $p(x)^2 \geq 0 \Rightarrow \boxed{\|p\|_2 \geq 0}$

$\|p_2\| = 0 \iff \boxed{p = 0}$

2) $\boxed{\|\alpha p\|_2} = \sqrt{\int_0^1 \alpha^2 p(x)^2\,dx} = \sqrt{\alpha^2 \int_0^1 p(x)^2\,dx} = |\alpha|\sqrt{\int_0^1 p(x)^2\,dx} \boxed{= |\alpha| \cdot \|p\|_2}$

3) $\|p+q\|_2^2 = \int_0^1 (p+q)^2\,dx = \int_0^1 p^2\,dx + 2\int_0^1 pq\,dx + \int_0^1 q^2\,dx \overset{\text{th. CS}}{\leq} \int_0^1 p^2\,dx + 2\sqrt{\int_0^1 p^2\,dx}\sqrt{\int_0^1 q^2\,dx} + \int_0^1 q^2\,dx =$

$= \|p\|_2^2 + 2\|p\|_2 \cdot \|q\|_2 + \|q\|_2^2 = (\|p\|_2 + \|q\|_2)^2$

$\Rightarrow \boxed{\|p+q\|_2 \leq \|p\|_2 + \|q\|_2}$

It's a norm on P_n.

4 $p_1(x) = x+1 \qquad p_2(x) = 1-x$

$\langle p_1, p_2 \rangle = \int_0^1 (x+1)(1-x)\,dx = \int_0^1 (1-x^2)\,dx = x - \frac{x^3}{3}\Big|_0^1 = \left(1-\frac{1}{3}\right) - \left(0-\frac{0}{3}\right) = \frac{2}{3}$

$\|p_1\|^2 = \int_0^1 (x+1)^2\,dx = \frac{(x+1)^3}{3}\Big|_0^1 = \frac{(1+1)^3}{3} - \frac{(0+1)^3}{3} = \frac{7}{3}$

$\|p_2\|^2 = \int_0^1 (1-x)^2\,dx = -\frac{(1-x)^3}{3}\Big|_0^1 = \frac{-(1-1)^3}{3} + \frac{(1-0)^3}{3} = \frac{1}{3}$

$\cos\theta = \frac{\frac{2}{3}}{\sqrt{\frac{7}{3}}\sqrt{\frac{1}{3}}} = \frac{2}{\sqrt{7}}$

$\boxed{\theta = \arccos\left(\frac{2}{\sqrt{7}}\right)}$

c) 1) $\boxed{\max\{|x_1|, |x_2|\} \geq 0}$, because $|x_1| \geq 0$ and $|x_2| \geq 0$

$\max\{|x_1|, |x_2|\} = 0 \Rightarrow |x_1| = |x_2| = 0 \Rightarrow x_1 = x_2 = 0 \Rightarrow \boxed{\|x\|_\infty = 0}$

2) $\boxed{\|\alpha x\|_\infty} = \max\{|\alpha x_1|, |\alpha x_2|\} = \max\{|\alpha|\cdot|x_1|, |\alpha|\cdot|x_2|\} = |\alpha| \cdot \max\{|x_1|, |x_2|\} = \boxed{|\alpha| \cdot \|x\|_\infty}$

3) $|x_i| \leq \|x\|_\infty, \quad |y_i| \leq \|y\|_\infty$

$|x_i + y_i| \leq |x_i| + |y_i| \leq \|x\|_\infty + \|y\|_\infty$ for $\forall i \Rightarrow \boxed{\|x+y\|_\infty \leq \|x\|_\infty + \|y\|_\infty}$

2 $x = (x_1, x_2) \qquad y = (y_1, y_2) \qquad \langle x, y \rangle = x_1 y_1 - (x_1 y_2 + x_2 y_1) + 2 x_2 y_2$

a) symmetry: $\boxed{\langle y, x \rangle} = y_1 x_1 - (y_1 x_2 + y_2 x_1) + 2 y_2 x_2 \boxed{= \langle x, y \rangle}$

linearity: $\boxed{\langle \alpha x + \beta z, y \rangle} = (\alpha x_1 + \beta z_1) y_1 - \big((\alpha x_1 + \beta z_1) y_2 + (\alpha x_2 + \beta z_2) y_1\big) + 2(\alpha x_2 + \beta z_2) y_2 =$

$= \alpha\big(x_1 y_1 - (x_1 y_2 + x_2 y_1) + 2 x_2 y_2\big) + \beta\big(z_1 y_1 - (z_1 y_2 + z_2 y_1) + 2 z_2 y_2\big) \boxed{= \alpha \langle x, y \rangle + \beta \langle z, y \rangle}$

positive definiteness: $\langle x, x \rangle = x_1^2 - (x_1 x_2 + x_2 x_1) + 2 x_2^2 = x_1^2 - 2 x_1 x_2 + 2 x_2^2 =$

$= x_1^2 - 2 x_1 x_2 + x_2^2 + x_2^2 = (x_1 - x_2)^2 + x_2^2 \geq 0$

if $x_1 - x_2 = 0$ and $x_2 = 0 \Rightarrow x_1 = x_2 = 0$

$\Rightarrow \boxed{\langle x, x \rangle > 0 \text{ for } x \neq 0}$.

It's an inner product

b) $\|x\| = \sqrt{\langle x, x \rangle} \boxed{= \sqrt{(x_1 - x_2)^2 + x_2^2}}$

c) $v = (1, 2), \quad \|v\| = \sqrt{(1-2)^2 + 2^2} = \sqrt{1+4} = \boxed{\sqrt{5}}$

d) $v = (1, 2), \quad w = (-1, 1)$

$$\cos\theta = \frac{\langle v, w \rangle}{\|v\| \cdot \|w\|} = \frac{-1\cdot 1 - (-1\cdot 2 + 1\cdot 1) + 2\cdot 2\cdot 1}{\sqrt{5} \cdot \sqrt{(-1-1)^2 + 1^2}} = \frac{-1 + 1 + 4}{\sqrt{5} \cdot \sqrt{5}} = \frac{4}{5}$$

$\boxed{\theta = \arccos\left(\frac{4}{5}\right)}$

5 $v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

(a) $\|v_1\|_1 = |1| + |2| = \boxed{3}$

$\|v_1\|_2 = \sqrt{1^2 + 2^2} = \boxed{\sqrt{5}}$

$\|v_1\|_\infty = \max\{|1|, |2|\} = \boxed{2}$

$\|v_2\|_1 = |-1| + |1| = \boxed{2}$

$\|v_2\|_2 = \sqrt{(-1)^2 + 1^2} = \boxed{\sqrt{2}}$

$\|v_2\|_\infty = \max\{|-1|, |1|\} = \boxed{1}$

(b) $d(v_1, v_2) = \|v_1 - v_2\|$

$v_1 - v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ – it's almost the same like v_1, so: $d_1 = \boxed{3}$

$d_2 = \boxed{\sqrt{5}}$

$d_\infty = \boxed{2}$